Neuromodulated Spike-Timing-Dependent Plasticity

A Three-Neuron Model for Spatial Credit Assignment

1 Circuit and Neural Dynamics

Two presynaptic neurons (pre_1, pre_2) project onto one postsynaptic LIF neuron through independent synapses of weight w_1 and w_2. Pre_1 is the *target* input whose activity is paired with reward; pre_2 is a *distractor* that fires at the same rate but is not reward-paired. Both fire independent Poisson spike trains.

1.1 Membrane Potential (LIF)

The postsynaptic neuron is a leaky integrate-and-fire (LIF) unit (Dayan & Abbott, 2001; Gerstner et al., 2014). Its membrane potential $V(t)$ obeys

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m\big(w_1 I_{s,1} + w_2 I_{s,2}\big) \tag{1}$$

where $\tau_m = R_m C_m$ is the membrane time constant, E_L the resting potential, and $I_{s,i}$ the synaptic current from synapse i. A spike fires when $V(t) = \theta$ with $dV/dt > 0$; the membrane resets to V_{reset} and integration is suspended for τ_{ref}.

1.2 Presynaptic Spike Trains

Each presynaptic neuron fires independent Poisson spikes:

$$\rho_i(t) = \sum_k \delta\big(t - t_i^k\big), \quad i \in \{1, 2\} \tag{2}$$

The postsynaptic spike train $\rho_{\text{post}(t)}$ is determined by Equation 1. Published three-factor STDP simulations universally use Poisson (not deterministic) presynaptic input (Florian, 2007; Frémaux et al., 2010; Izhikevich, 2007; Vasilaki et al., 2009).

1.3 Synaptic Current

Each synapse has its own current, driven by its presynaptic spike train filtered through an exponential kernel:

$$\frac{dI_{s,i}}{dt} = -\frac{I_{s,i}}{\tau_s}, \quad I_{s,i} \leftarrow I_{s,i} + 1 \text{at each pre}_i \text{ spike} \tag{3}$$

2 Three-Factor Plasticity

The plasticity rule belongs to the class of *three-factor learning rules* (Frémaux & Gerstner, 2016; Izhikevich, 2007). Standard STDP depends on pre–post spike correlations (two factors); three-factor rules gate this local signal with a global neuromodulatory factor M.

2.1 STDP Traces

Each synapse has a presynaptic trace x_i; the postsynaptic trace y_{post} is shared (Bi & Poo, 1998):

$$\tau_+ \frac{dx_i}{dt} = -x_i + \rho_i(t), \quad \tau_- \frac{dy_{\text{post}}}{dt} = -y_{\text{post}} + \rho_{\text{post}(t)} \tag{4}$$

2.2 Eligibility Trace

Coincident pre- and post-spikes create a per-synapse *eligibility trace* $E_i(t)$ (Gerstner et al., 2018; Izhikevich, 2007) that bridges millisecond spike timing and delayed reward:

$$\tau_e \frac{dE_i}{dt} = -E_i + \underbrace{\eta_+(w_{\max} - w_i)x_i\rho_{\text{post}}}_{\text{LTP}} - \underbrace{\eta_- w_i y_{\text{post}}\rho_i}_{\text{LTD}} \tag{5}$$

Soft-bound factors $(w_{\max} - w_i)$ and w_i implement *multiplicative STDP* (Song et al., 2000), preventing unbounded growth.

2.3 Firing-Rate Estimation

Smooth rate estimates are obtained by exponential filtering:

$$\tau_r \frac{dr_i}{dt} = -r_i + \rho_i(t), \quad \tau_r \frac{dr_{\text{post}}}{dt} = -r_{\text{post}} + \rho_{\text{post}(t)} \tag{6}$$

2.4 Weight Update

Each weight evolves under the same global modulation $M(t)$, gated by its own eligibility:

$$\frac{dw_i}{dt} = M(t) \cdot E_i(t) \tag{7}$$

Because M is global but E_i is synapse-specific, only the synapse with high eligibility at the moment of reward undergoes a large weight change. This is the mechanism of *spatial credit assignment*.

3 Neuromodulator Roles

The neuromodulator $M(t)$ determines *how* the reward signal R influences plasticity. Frémaux & Gerstner (Frémaux & Gerstner, 2016) (Eq. 14) identify four roles:

3.1 Covariance (RPE / Dopaminergic)

$$M = R - \overline{R}, \quad \tau_{\overline{R}} \frac{d\overline{R}}{dt} = -\overline{R} + R \tag{8}$$

A *reward prediction error*: phasic dopamine encodes "better than expected" vs "worse than expected" (Schultz et al., 1997). The baseline $\overline{R}$ adapts slowly ($\tau_{\overline{R}} = 5$ s), acting as a critic (Sutton & Barto, 1998). This is the default in R-STDP (Frémaux & Gerstner, 2016) (Section 4.3, Eq. 7).

3.2 Gated Hebbian

$$M = R \tag{9}$$

Reward directly gates plasticity with no baseline subtraction.

3.3 Surprise / Novelty

$$M = \left(r_{\text{post}} - \overline{R}\right)^2, \quad \tau_{\overline{R}} \frac{d\overline{R}}{dt} = -\overline{R} + r_{\text{post}} \tag{10}$$

Here $\overline{R}$ tracks the *firing rate*, not reward. The surprise signal is always non-negative; LTP vs LTD direction is determined by E_i alone. Models noradrenergic or cholinergic modulation (Yu & Dayan, 2005).

3.4 Constant (Non-Modulated STDP)

$$M = 1 \tag{11}$$

Standard two-factor STDP. Useful as a baseline for isolating the effect of neuromodulation.

4 Reward Signals

The reward signal $R(t)$ defines *what* the system is rewarded for. In the three-factor STDP literature, R is always an external, task-based signal (Florian, 2007; Frémaux & Gerstner, 2016; Izhikevich, 2007; Legenstein et al., 2008).

4.1 Target Rate (Self-Supervisory Demonstration)

$$R = -\left(r_{\text{post}} - r_{\text{target}}\right)^2 \tag{12}$$

The postsynaptic rate is penalised for deviating from a target. This is self-supervisory (not from the literature) but useful for verifying the ODE machinery.

4.2 Biofeedback

$$R(t) = d(t), \quad d(t) = \begin{cases} R_{\text{amount}} \cdot e^{-(t - t_{\text{delivery}})/\tau_d} & \text{if reward delivered at } t_{\text{delivery}} \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

Every postsynaptic spike schedules a reward pulse at $t^k_{\text{post}} + \Delta_{\text{reward}}$. This is the Legenstein et al. (Legenstein et al., 2008) biofeedback paradigm. It demonstrates *temporal* credit assignment but not *spatial*: with two synapses onto the same neuron, both inputs are directly upstream and both strengthen.

4.3 Contingent Reward (Izhikevich Paradigm)

The same delayed pulse mechanism as biofeedback, but reward is scheduled **only when pre₁ fires within a coincidence window Δ_c before a post-spike**:

$$\text{schedule reward at } t + \Delta_{\text{reward}} \quad \text{iff} \quad \exists t^k_1 \in [t - \Delta_c, t] \tag{14}$$

where t is the post-spike time and t_1^k is the most recent pre_1 spike. This is the "gated-Hebbian" paradigm described by Frémaux & Gerstner (Frémaux & Gerstner, 2016) (Eq. 10) and used by Izhikevich (Izhikevich, 2007):

$$\langle \dot{w}_i \rangle = \text{Cov}(R, H_i) + \langle R \rangle \langle H_i \rangle \tag{15}$$

where $H_i = H(\text{pre}_i, \text{post})$ is the Hebbian coincidence term for synapse i. Because reward is correlated with $\text{pre}_1 \to \text{post}$ coincidences:

- $\text{Cov}(R, H_1) > 0$: synapse 1 strengthens.
- $\text{Cov}(R, H_2) \approx 0$: synapse 2 sees no systematic correlation.

The $\langle R \rangle \langle H_i \rangle$ bias drives both weights downward (since $\langle H_i \rangle < 0$ under typical STDP windows), but the positive covariance for synapse 1 overcomes this drift. Reward acts as a *binary gating signal* that switches from general depression to selective potentiation of the target synapse.

5 Complete System of Equations

State vector: $\boldsymbol{x}(t) = \big(V, y_{\text{post}}, r_{\text{post}}, \overline{R}, I_{s,1}, x_1, E_1, r_1, w_1, I_{s,2}, x_2, E_2, r_2, w_2\big) \in \mathbb{R}^{14}$.

5.0.1 Inter-spike dynamics

$$\begin{aligned}
\tau_m \frac{dV}{dt} &= -(V - E_L) + R_m\big(w_1 I_{s,1} + w_2 I_{s,2}\big) \\
\frac{dy_{\text{post}}}{dt} &= -y_{\text{post}}/\tau_- \\
\frac{dr_{\text{post}}}{dt} &= -r_{\text{post}}/\tau_r \\
\tau_{\overline{R}} \frac{d\overline{R}}{dt} &= -\overline{R} + T_{\text{neuromod}} \\
\frac{dI_{s,i}}{dt} &= -I_{s,i}/\tau_s \\
\frac{dx_i}{dt} &= -x_i/\tau_+ \\
\frac{dE_i}{dt} &= -E_i/\tau_e \\
\frac{dr_i}{dt} &= -r_i/\tau_r \\
\frac{dw_i}{dt} &= M \cdot E_i
\end{aligned} \tag{16}$$

where T_{neuromod} is the baseline tracking target: R for covariance and gated, r_{post} for surprise, $\overline{R}$ for constant.

5.0.2 Pre-synaptic spike at t_i^k

$$I_{s,i} \leftarrow I_{s,i} + 1, \quad x_i \leftarrow x_i + 1, \quad r_i \leftarrow r_i + 1 \tag{17}$$

$$E_i \leftarrow E_i - \eta_- w_i y_{\text{post}(t_i^{k-})} \quad \text{(LTD eligibility)} \tag{18}$$

5.0.3 Post-synaptic spike at t_{post}^k

$$V \leftarrow V_{\text{reset}}, \quad y_{\text{post}} \leftarrow y_{\text{post}} + 1, \quad r_{\text{post}} \leftarrow r_{\text{post}} + 1 \tag{19}$$

$$E_i \leftarrow E_i + \eta_+(w_{\max} - w_i)x_i\left(t_{\text{post}}^{k-}\right) \quad \text{(LTP eligibility, both synapses)} \tag{20}$$

5.0.4 Reward scheduling (contingent mode)

If pre_1 fired within Δ_c before the post-spike, schedule a global reward pulse at $t_{\text{post}}^k + \Delta_{\text{reward}}$.

5.0.5 Summary table

Neuromodulator	M	$\overline{R}$ tracks	Source
Covariance	$R - \overline{R}$	R	(Frémaux & Gerstner, 2016) Eq. 7
Gated	R	R	(Frémaux & Gerstner, 2016) Eq. 14
Surprise	$\left(r_{\text{post}} - \overline{R}\right)^2$	r_{post}	(Yu & Dayan, 2005)
Constant	1	—	Two-factor baseline

6 Numerical Methods

6.1 Hybrid Integration Scheme

Smooth dynamics are integrated by Euler or classical RK4 between spike events. Discrete events (pre_1/pre_2/post spikes) apply instantaneous jumps to the state vector.

For the RK4 path, threshold crossing detection uses linear interpolation: given V_0 and $V_1 = V(t + \Delta t)$ from a trial step, the crossing fraction is

$$f = \frac{\theta - V_0}{V_1 - V_0} \tag{21}$$

The timestep is split: integrate to $t + f\Delta t$, apply the spike, integrate the remainder. Spike-time accuracy is $\mathcal{O}(\Delta t^2)$.

6.2 RK4 Convergence

Spike discontinuities violate the smoothness assumption underlying classical convergence theorems, reducing RK4′s effective order:

Variable	Euler (expect 1)	RK4 (expect 4)	Note
V (mV)	0.20	1.57	Degraded by spike discontinuities
w	1.79	1.25	Slow variable; averaging effects

7 Spatial Credit Assignment

The core result of the three-neuron model. Both synapses receive the same global M, but only synapse 1 has high E_1 when the delayed reward arrives.

7.1 Mechanism

1. Pre_1 fires before a post-spike $\rightarrow x_1$ is high $\rightarrow E_1$ receives a large LTP jump.
2. The post-spike triggers a contingent reward check: pre_1 fired within $\Delta_c \rightarrow$ reward scheduled at $t + \Delta_{\text{reward}}$.
3. The STDP traces (x_i, y_{post}) decay within ~ 20 ms, but E_1 persists ($\tau_e = 500$ ms).
4. When the reward pulse arrives (~ 1 s later), $R > 0 \rightarrow M > 0 \rightarrow dw_1/dt = M \cdot E_1 > 0$.
5. Pre_2 fires independently. Its eligibility E_2 is sometimes nonzero at reward time, but there is no systematic correlation $\rightarrow \langle M \cdot E_2 \rangle \approx 0$.

Over many reward events, w_1 grows while w_2 remains flat or drifts downward.

7.2 Timescale Hierarchy

Variable	Dominant dynamics	Timescale
V	Spikes + subthreshold integration	$\tau_m = 20$ ms
$I_{s,i}$	Impulse + fast decay	$\tau_s = 5$ ms
x_i, y_{post}	Impulse + moderate decay	$\tau_\pm = 20$ ms
E_i	STDP jumps + slow decay	$\tau_e = 500$ ms
r_i, r_{post}	Impulse + slow decay	$\tau_r = 500$ ms
$\overline{R}$	Very slow tracking	$\tau_{\overline{R}} = 5$ s
w_i	Slow drift via $M \cdot E_i$	Emergent; seconds

The eligibility trace ($\tau_e = 500$ ms) is the critical bridge. It outlives the STDP traces (20 ms) but decays before the next reward (~ 1 s). This window is what allows the delayed reward to selectively read out which synapse was causal (Frémaux et al., 2010).

8 Parameters

Parameter	Description	Value	Source
τ_m	Membrane time constant	20 ms	(Dayan & Abbott, 2001)
E_L	Resting potential	-65 mV	(Dayan & Abbott, 2001)
R_m	Membrane resistance	50 MΩ	(Dayan & Abbott, 2001)
θ	Spike threshold	-50 mV	(Gerstner et al., 2014)
V_{reset}	Reset potential	-70 mV	(Gerstner et al., 2014)
τ_{ref}	Refractory period	3 ms	(Gerstner et al., 2014)
τ_s	Synaptic decay	5 ms	(Dayan & Abbott, 2001)
τ_+, τ_-	STDP windows	20 ms	(Bi & Poo, 1998)
τ_r	Rate trace decay	500 ms	(Gerstner et al., 2014)
τ_e	Eligibility decay	500 ms	(Frémaux & Gerstner, 2016)
$\tau_{\overline{R}}$	Reward baseline	5 s	(Frémaux & Gerstner, 2016)
η_+, η_-	Learning rates	10^{-4}	(Song et al., 2000)
$w_{\max}$	Max weight	10	—
$w_{1,0}, w_{2,0}$	Initial weights	2.0	—

Parameter	Description	Value	Source
r_{pre}	Poisson firing rate (each)	20 Hz	(Frémaux et al., 2010)
Δ_{reward}	Reward delay	1.0 s	(Izhikevich, 2007)
R_{amount}	Reward pulse amplitude	1.0	(Izhikevich, 2007)
τ_d	Reward pulse decay	200 ms	—
Δ_c	Coincidence window	20 ms	(Izhikevich, 2007)

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